Elementary Analysis

Section 24: Uniform Convergence

Exercises 24.1–24.17, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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24 Uniform convergence

Before starting. Pointwise convergence means fixing one input x first and then letting n grow. Uniform convergence means finding one index that works for *all* inputs at once. We shall repeatedly estimate the largest error

$$E_n = \sup_{x \in S} |f_n(x) - f(x)|.$$

Uniform convergence is equivalent to $E_n \to 0$; Exercise 24.12 proves this. To disprove uniform convergence, it is enough to find a positive number ε_0 and inputs $x_n \in S$, for arbitrarily large n, with $|f_n(x_n) - f(x_n)| \geq \varepsilon_0$. The input is allowed to change with n. That is precisely the obstacle uniform convergence must overcome.

24.1 Exercise 24.1: One bound for every oscillation

What we must show. Prove uniform convergence to zero for $f_n(x) = (1 + 2\cos^2(nx))/\sqrt{n}$ on $\mathbb{R}$.

Plan. We need not follow the oscillations: the cosine is always between -1 and 1.

Bound the numerator independently of the input.

Since $0 \leq \cos^2(nx) \leq 1$, we have

$$|f_n(x) - 0| \leq \frac{3}{\sqrt{n}} \qquad (x \in \mathbb{R}).$$

Given $\varepsilon > 0$, choose an integer $N > (3/\varepsilon)^2$. For every $n > N$ and every real x, the last bound is smaller than ε. The choice of N depends only on ε.

Conclusion. $f_n \to 0$ uniformly on $\mathbb{R}$.

24.2 Exercise 24.2: A small slope on bounded and unbounded domains

What we must show. Analyze $f_n(x) = x/n$ on $[0, \infty)$.

(a) Fix the input before passing to the limit.

For fixed $x \geq 0$, the numerator is constant and $n \to \infty$, so $f_n(x) \to 0$. Thus the pointwise limit is $f(x) = 0$.

(b) Control all inputs in $[0, 1]$.

There $|f_n(x)| \leq 1/n$. Given $\varepsilon > 0$, any integer $N > 1/\varepsilon$ works for all $x \in [0, 1]$ and $n > N$.

(c) Let the input escape along with the index.

On $[0, \infty)$ take $x_n = n$. Then $f_n(x_n) = 1$, so the error never falls below $1/2$. This contradicts the requirement of uniform convergence.

Conclusion. The limit is zero everywhere; convergence is uniform on $[0, 1]$ but not on $[0, \infty)$.

24.3 Exercise 24.3: A transition concentrated near one

What we must show. Analyze $f_n(x) = 1/(1+x^n)$ for $x \geq 0$.

(a) Separate the three possible behaviors of a power.

If $0 \leq x < 1$, then $x^n \to 0$, giving limit 1. At $x = 1$ every term is $1/2$. If $x > 1$, then $x^n \to \infty$, giving limit 0. Thus

$$f(x) = \begin{cases} 1 & 0 \leq x < 1, \\ 1/2 & x = 1, \\ 0 & x > 1. \end{cases}$$

(b) Test uniform convergence on $[0,1]$.

Each f_n is continuous there because its denominator is positive. The limit is not continuous at 1: its values from the left are 1, whereas $f(1) = 1/2$. The uniform-limit theorem for continuous functions therefore excludes uniform convergence on $[0,1]$.

(c) Use restriction to a smaller domain.

Uniform convergence on $[0,\infty)$ would imply uniform convergence on its subset $[0,1]$, with exactly the same error bound. Part (b) rules that out.

Conclusion. The piecewise function above is the pointwise limit, and neither of the two requested uniform-convergence assertions holds.

24.4 Exercise 24.4: The complementary transition

What we must show. Analyze $f_n(x) = x^n/(1+x^n)$ on $[0,\infty)$.

(a) Use an exact identity.

We have $f_n(x) = 1 - 1/(1+x^n)$. The three cases of the preceding exercise therefore give

$$f(x) = \begin{cases} 0 & 0 \leq x < 1, \\ 1/2 & x = 1, \\ 1 & x > 1. \end{cases}$$

(b) Notice the endpoint jump.

The f_n are continuous on $[0,1]$, but f is not continuous at 1. Consequently convergence cannot be uniform there, by Theorem 24.3.

(c) Restrict any proposed larger-domain convergence.

Uniform convergence on $[0,\infty)$ would restrict to uniform convergence on $[0,1]$, contradicting part (b).

Conclusion. The displayed f is the limit; convergence is uniform on neither $[0,1]$ nor $[0,\infty)$.

24.5 Exercise 24.5: A denominator containing both the index and a power

What we must show. Analyze $f_n(x) = x^n/(n + x^n)$ for $x \geq 0$.

(a) Find the pointwise limit.

For $0 \leq x \leq 1$, we have $0 \leq f_n(x) \leq 1/n$, so the limit is zero. For fixed $x > 1$, divide numerator and denominator by x^n:

$$f_n(x) = \frac{1}{1 + n/x^n} \longrightarrow 1.$$

Here $n/x^n \to 0$ because exponential growth dominates the linear factor; for example, the ratio of successive terms tends to $1/x < 1$. Thus $f(x) = 0$ on $[0, 1]$ and $f(x) = 1$ on $(1, \infty)$.

(b) Obtain a common bound on $[0, 1]$.

The estimate $|f_n(x) - f(x)| \leq 1/n$ works simultaneously for every input in that interval. Hence convergence is uniform there.

(c) Use the jump at an interior point of the larger domain.

Every f_n is continuous on $[0, \infty)$, but the pointwise limit jumps at 1. The uniform-limit theorem excludes uniform convergence on the larger domain.

Conclusion. Uniform convergence holds on $[0, 1]$, and fails on $[0, \infty)$.

24.6 Exercise 24.6: Shifted squares

What we must show. Find the limit of $f_n(x) = (x - 1/n)^2$ on $[0, 1]$ and decide uniform convergence.

(a) Expand to identify the limit.

For fixed x, $f_n(x) = x^2 - 2x/n + 1/n^2 \to x^2$. Thus $f(x) = x^2$.

(b) Bound the entire error, rather than just each separate input.

For $0 \leq x \leq 1$,

$$|f_n(x) - x^2| = \left| -\frac{2x}{n} + \frac{1}{n^2} \right| \leq \frac{2}{n} + \frac{1}{n^2} \leq \frac{3}{n}.$$

Given $\varepsilon > 0$, choose $N > 3/\varepsilon$. This bound is then less than ε for every $x \in [0, 1]$ and $n > N$.

Conclusion. $f_n \to (x \mapsto x^2)$ uniformly on $[0, 1]$.

24.7 Exercise 24.7: Subtracting a power

What we must show. Analyze $f_n(x) = x - x^n$ on $[0, 1]$.

(a) Check the endpoint separately.

For $0 \leq x < 1$, $x^n \to 0$, so the limit is x. At $x = 1$, $f_n(1) = 0$ for every n. Hence

$$f(x) = \begin{cases} x & 0 \leq x < 1, \\ 0 & x = 1. \end{cases}$$

(b) Apply the continuity obstruction.

Each f_n is a polynomial and therefore continuous on $[0,1]$. The limit is discontinuous at 1, because its left-hand limit is 1 and its value there is zero. By Theorem 24.3 the convergence is not uniform.

Conclusion. There is pointwise convergence to the displayed function, but no uniform convergence on $[0,1]$.

24.8 Exercise 24.8: Partial geometric sums at their divergent endpoint

What we must show. Analyze $f_n(x) = \sum_{k=0}^{n} x^k$ on $[0,1]$.

(a) Evaluate the finite sum where its formula is valid.

For $0 \le x < 1$,

$$f_n(x) = \frac{1 - x^{n+1}}{1 - x} \longrightarrow \frac{1}{1 - x}.$$

At $x = 1$ there are $n + 1$ summands equal to 1, so $f_n(1) = n + 1 \to +\infty$. Thus there is *no real-valued pointwise limit on all of* $[0,1]$. An infinite limit at a point is not a real-valued limit function in Definition 24.1.

(b) Uniform convergence would imply pointwise convergence.

Since real-valued pointwise convergence already fails at 1, uniform convergence on $[0,1]$ is impossible.

Conclusion. Convergence occurs pointwise only on $[0,1)$, to $1/(1-x)$; it does not occur pointwise, or uniformly, on the full requested interval.

24.9 Exercise 24.9: A moving peak whose integral vanishes

What we must show. Study $f_n(x) = nx^n(1-x)$ on $[0,1]$.

(a) Fix an input.

At $x = 0$ and $x = 1$ the function is zero. For $0 < x < 1$, the ratio of successive terms of nx^n is $x(1 + 1/n) \to x < 1$, so $nx^n \to 0$. Multiplication by the fixed number $1 - x$ gives $f_n(x) \to 0$. Thus $f = 0$ everywhere on $[0,1]$.

(b) Find a peak that does not shrink to zero.

Differentiating gives

$$f_n'(x) = nx^{n-1}(n - (n+1)x).$$

The sign changes from positive to negative at $x_n = n/(n+1)$, the maximum point. Its value is

$$f_n(x_n) = \frac{n}{n+1}\left(\frac{n}{n+1}\right)^n = \left(\frac{n}{n+1}\right)^{n+1} \longrightarrow e^{-1} > 0.$$

The last limit follows from $(1 + 1/n)^n \to e$. Because the maximum error does not approach zero, convergence is not uniform.

(c) Compute the integrals directly.

Despite the preceding failure,

$$\int_0^1 f_n(x)\,dx = n\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{n}{(n+1)(n+2)} \longrightarrow 0 = \int_0^1 f(x)\,dx.$$

Conclusion. The limit is zero, convergence is not uniform, and the integrals nevertheless converge to the integral of the limit. Uniform convergence is a sufficient condition for this interchange, not a necessary one.

24.10 Exercise 24.10: Adding uniform limits, and the danger with products

What we must show. Prove the sum rule and examine its product analogue.

(a) Split the desired error into two smaller errors.

Let $\varepsilon > 0$. Uniform convergence supplies N_1, N_2 such that $|f_n(x) - f(x)| < \varepsilon/2$ for $n > N_1$ and $|g_n(x) - g(x)| < \varepsilon/2$ for $n > N_2$, in each case for all $x \in S$. For $n > \max\{N_1, N_2\}$ the triangle inequality gives

$$|(f_n + g_n)(x) - (f + g)(x)| \le |f_n(x) - f(x)| + |g_n(x) - g(x)| < \varepsilon.$$

This is uniform convergence of the sums.

(b) Test the product claim before accepting it.

Products need not converge uniformly: on $S = \mathbb{R}$, let $f_n(x) = x$ and $g_n(x) = 1/n$. Their limits are $f(x) = x$ and $g(x) = 0$, both uniform. But $f_n(x)g_n(x) = x/n$ has value 1 at $x = n$, so it fails to converge uniformly to $fg = 0$. A small error multiplied by an unbounded function can become large; this is the missing issue in a naive product proof.

Conclusion. Sums preserve uniform convergence; unrestricted products do not.

24.11 Exercise 24.11: The product counterexample in full

What we must show. Verify every part of the example $f_n(x) = x$, $g_n(x) = 1/n$ on $\mathbb{R}$.

(a) Compute the two errors separately.

With $f(x) = x$ we have $|f_n(x) - f(x)| = 0$ for every n, x. With $g(x) = 0$ we have $|g_n(x) - g(x)| = 1/n$, independently of x. The first error is already zero, and the second tends to zero, so both convergences are uniform.

(b) Compute the product error at a suitable input.

For each n, use $x_n = n$:

$$|f_n(x_n)g_n(x_n) - f(x_n)g(x_n)| = |n/n - 0| = 1.$$

No common index can make all these errors smaller than $1/2$.

Conclusion. $f_n \to f$ and $g_n \to g$ uniformly, but $f_n g_n \not\to fg$ uniformly on $\mathbb{R}$.

24.12 Exercise 24.12: The supremum-error characterization

What we must show. Prove that $f_n \to f$ uniformly on S if and only if $E_n = \sup_{x \in S} |f_n(x) - f(x)| \to 0$. For an empty domain the assertion is vacuous; take its maximum error to be zero. For a nonempty domain, E_n is initially allowed to be $+\infty$.

Uniform convergence implies vanishing suprema.

Given $\varepsilon > 0$, use uniform convergence with $\varepsilon/2$. For sufficiently large n, all the errors are smaller than $\varepsilon/2$, so their supremum satisfies $0 \le E_n \le \varepsilon/2 < \varepsilon$. Notice the non-strict inequality at the supremum: a least upper bound need not be attained. The extra factor $1/2$ avoids any difficulty.

Vanishing suprema imply uniform convergence.

Given $\varepsilon > 0$, choose N with $E_n < \varepsilon$ for all $n > N$. Every individual error is at most its supremum, so $|f_n(x) - f(x)| \le E_n < \varepsilon$ for all $x \in S$ and $n > N$. This is exactly the definition of uniform convergence.

Conclusion. The two conditions are equivalent. The supremum is over the input variable first; the limit is then taken over the index.

24.13 Exercise 24.13: Uniform limits of uniformly continuous functions

What we must show. Show that a uniform limit f of uniformly continuous f_n on (a, b) is uniformly continuous there.

Plan. Approximate f by one fixed f_n, and let that function control nearby inputs. There will be three errors, each allocated $\varepsilon/3$.

Choose one approximating function before choosing the inputs.

Given $\varepsilon > 0$, uniform convergence supplies an index m with $|f_m(t) - f(t)| < \varepsilon/3$ for all $t \in (a, b)$. Uniform continuity of this particular f_m supplies $\delta > 0$ such that $|f_m(x) - f_m(y)| < \varepsilon/3$ whenever $x, y \in (a, b)$ and $|x - y| < \delta$.

Join the three estimates.

For such x, y,

$$|f(x) - f(y)| \le |f(x) - f_m(x)| + |f_m(x) - f_m(y)| + |f_m(y) - f(y)| < \varepsilon.$$

Neither m nor δ was chosen using x or y, so this is uniform continuity, rather than merely continuity at a single point.

Conclusion. The uniform limit is uniformly continuous. The same proof works on any subset of $\mathbb{R}$, not just an interval.

24.14 Exercise 24.14: A narrow peak near zero

What we must show. Analyze $f_n(x) = nx/(1 + n^2x^2)$ and the proposed limit $f = 0$.

(a) Fix x and distinguish zero from nonzero.

At $x = 0$ every term is zero. If $x \neq 0$, dividing by n^2x^2 gives

$$f_n(x) = \frac{1/(nx)}{1 + 1/(n^2x^2)} \longrightarrow 0.$$

Thus pointwise convergence holds on $\mathbb{R}$.

(b) Test the shrinking input $x_n = 1/n$ on $[0, 1]$.

At this input, $f_n(x_n) = 1/2$. Therefore uniform convergence to zero fails on $[0, 1]$. Each fixed positive input eventually lies past the peak, but the peak itself remains of height $1/2$.

(c) Keep a fixed positive distance from zero.

For $x \geq 1$,

$$0 \leq \frac{nx}{1 + n^2x^2} \leq \frac{nx}{n^2x^2} = \frac{1}{nx} \leq \frac{1}{n}.$$

The upper bound tends to zero independently of x, proving uniform convergence on $[1, \infty)$.

Conclusion. Pointwise convergence holds everywhere; uniform convergence fails on $[0, 1]$ and holds on $[1, \infty)$.

24.15 Exercise 24.15: A boundary layer with limiting height one

What we must show. Analyze $f_n(x) = nx/(1 + nx)$ on $[0, \infty)$.

(a) Identify the limit.

At zero the value is zero. If $x > 0$, then $f_n(x) = 1 - 1/(1 + nx) \to 1$. Thus $f(0) = 0$ and $f(x) = 1$ for $x > 0$.

(b) Inspect the interval containing the boundary layer.

The continuous f_n have a discontinuous pointwise limit at zero, so convergence cannot be uniform on $[0, 1]$. Explicitly, $x_n = 1/n$ gives $f_n(x_n) = 1/2$ and $f(x_n) = 1$, an error of $1/2$.

(c) Bound the error on $[1, \infty)$.

For $x \geq 1$ the limit is 1, and

$$|f_n(x) - 1| = \frac{1}{1 + nx} \leq \frac{1}{1 + n} \longrightarrow 0.$$

This is an input-independent bound and hence proves uniform convergence.

Conclusion. The limit jumps from 0 to 1 at zero; convergence is not uniform on $[0, 1]$ but is uniform on $[1, \infty)$.

24.16 Exercise 24.16: A boundary layer with an unbounded limiting function

What we must show. Analyze $f_n(x) = nx/(1 + nx^2)$ for $x \geq 0$.

(a) Compute the pointwise limit.

At zero the value is zero. For $x > 0$, division by n gives $f_n(x) = x/(1/n + x^2) \to 1/x$. Thus $f(0) = 0$ and $f(x) = 1/x$ for $x > 0$.

(b) Show failure of uniform convergence on $[0, 1]$.

One obstruction is the discontinuity, indeed unboundedness, of the limit near zero. More directly, at $x_n = 1/\sqrt{n}$,

$$f_n(x_n) = \frac{\sqrt{n}}{2}, \qquad f(x_n) = \sqrt{n}, \qquad |f_n(x_n) - f(x_n)| = \frac{\sqrt{n}}{2}.$$

These errors grow, so they certainly do not tend uniformly to zero.

(c) Estimate the error away from the singular endpoint.

For $x \geq 1$,

$$\left| \frac{nx}{1 + nx^2} - \frac{1}{x} \right| = \frac{1}{x(1 + nx^2)} \leq \frac{1}{1 + n} \longrightarrow 0.$$

Conclusion. The displayed piecewise f is the limit; convergence fails to be uniform on $[0, 1]$ and is uniform on $[1, \infty)$.

24.17 Exercise 24.17: Passing to the limit while the input also moves

What we must show. If $f_n \to f$ uniformly on $[a, b]$, the f_n are continuous, and $x_n \to x$ in that interval, prove $f_n(x_n) \to f(x)$.

First secure continuity of the limiting function.

By Theorem 24.3, the uniform limit f is continuous on $[a, b]$. Also $x \in [a, b]$, since this interval is closed. Thus continuity gives $f(x_n) \to f(x)$.

Separate the changing function from the changing input.

For every n,

$$|f_n(x_n) - f(x)| \leq |f_n(x_n) - f(x_n)| + |f(x_n) - f(x)|.$$

Given $\varepsilon > 0$, uniform convergence makes the first term smaller than $\varepsilon/2$ for all sufficiently large n, *even at the moving input* x_n. Continuity and $x_n \to x$ make the second term smaller than $\varepsilon/2$ for all sufficiently large n. Taking n beyond both thresholds proves the desired limit.

Conclusion. $f_n(x_n) \to f(x)$. Uniform convergence is what allows us to substitute a different input at every index.